Amagi Architectural Class Definition

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1. Scope and Purpose

This document establishes the **formal definition of the Amagi architectural class** and its **irreducible architectural invariants**.

It serves as the **authoritative, non-negotiable reference** for determining whether a system constitutes an authentic Amagi-class architecture.

This document exists to:

- define the Amagi architectural class as a distinct and bounded category;
- prevent mischaracterization, dilution, or structural imitation of the Amagi Framework;
- provide a stable institutional reference independent of any specific implementation;
- serve as public prior art and a governance anchor for certification, verification, and regulatory alignment.

This document defines an architectural class **independently of jurisdiction, application domain, or regulatory regime**.

This document does **not** disclose trade secrets, confidential implementation details, or domain-specific hard intellectual property.

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2. Definition of the Amagi Architectural Class

Amagi is defined as an **architectural class**, not as an implementation pattern, product, feature set, software framework, or deployment model.

Membership in the Amagi architectural class is determined **exclusively** by satisfaction of the architectural invariants defined in this document.

Similarity in design intent, functional outcome, claimed purpose, or regulatory alignment does **not** constitute Amagi-class equivalence.

Amagi-class membership is a binary condition: systems either satisfy the defined invariants in full or do not belong to the Amagi architectural class.

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3. Irreducible Architectural Invariants

An Amagi-class system **shall** satisfy **all** of the following invariants. These invariants are **necessary and indivisible conditions** for Amagi-class membership.

AMAGI-INVARIANT-001: Physical Non-Bypassability

Control enforcement **shall** be guaranteed by physical construction and hardware isolation. Software convention, policy, runtime checks, virtualization, or best-effort enforcement mechanisms alone are insufficient.

AMAGI-INVARIANT-002: Irreducible Trusted Computing Base

A minimal Security Council Core (SCC) **shall** constitute the sole trusted control plane.

The SCC **shall** be structurally minimal, isolated, and amenable to formal verification.

AMAGI-INVARIANT-003: Hardware-Enforced Separation of Control and Cognition

Cognitive components **shall** be treated as untrusted and **shall not** possess direct authority over consequential actions, actuation paths, or state transitions.

Authority separation **shall** be enforced structurally and physically, not contractually or procedurally.

AMAGI-INVARIANT-004: Immutability of Critical Policy Structures

Safety-critical policy, access graphs, and enforcement logic **shall** be immutable post-deployment or fused into hardware or one-time-programmable memory.

Post-deployment modification, self-modification, or dynamic rewriting of these structures is not permitted.

AMAGI-INVARIANT-005: Complete Mediation

All consequential actions, actuation paths, and state transitions **shall** transit exclusively through the trusted control plane.

No side channels, bypass paths, or auxiliary control mechanisms may exist outside this mediation boundary.

AMAGI-INVARIANT-006: Complete Audit Integrity

Audit records **shall** be cryptographically protected, append-only, and structurally inaccessible to cognitive components.

Cognitive systems **shall not** be capable of modifying, suppressing, or selectively influencing audit visibility.

Invariant Boundary Condition

Removal, weakening, optionalization, substitution, or conditional application of any invariant defined above results in a system that is **not** Amagi-class.

Partial compliance is not defined.

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4. Irreducibility Statement

The Amagi architectural class is **irreducibly complex**.

The defined invariants are not design preferences, implementation guidelines, or optimization targets.
They arise directly from the requirement to enforce safety, authority limitation, and compliance as **structural properties**, rather than procedural assurances or behavioral expectations.

Any simplification that removes, relaxes, externalizes, or defers enforcement of an invariant collapses the architecture into a weaker and categorically distinct class.

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5. Relationship to Open Publication

The Amagi Open Specification and related publications are released as public prior art to enable independent analysis and to prevent enclosure through patents or proprietary claims.

Open publication does **not** imply permissive implementation rights, waiver of architectural constraints, or automatic Amagi-class equivalence.

Claims of Amagi compatibility require demonstrable structural equivalence to the invariants defined in this document.

5.1 Relationship to Regulatory and Standards Frameworks

This document is intended to be compatible with, but not limited to, emerging regulatory and standards frameworks governing high-risk and safety-critical systems, including AI governance, critical infrastructure protection, and formally assured computing.

References to such frameworks are **non-normative** and do not constrain, modify, or supersede the architectural invariants defined herein.

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6. Custodianship and Institutional Role

AGIAM acts as the architectural custodian of the Amagi architectural class.

Custodianship includes:

- maintenance of the canonical definition of Amagi-class invariants;
- definition of structural equivalence and verification criteria;
- preservation of architectural continuity across versions and implementations.

Custodianship does **not** imply ownership of intelligence, models, data, applications, or outcomes built within or around Amagi-class systems.

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7. Relationship to Trade Secrets and Licensing

This document:

- does not grant access to confidential implementation knowledge;
- does not supersede licensing requirements for commercial deployment;
- does not disclose domain-specific optimization, calibration, verification environments, or manufacturing processes.

Implementation-level knowledge may remain protected under applicable trade secret, licensing, and contractual regimes.

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8. Concluding Statement

The Amagi architectural class does not attempt to predict, align, or regulate intent.

It bounds system capability through architecture.

This document constitutes a **public, timestamped declaration of the architectural boundary that defines Amagi-class systems**.
